GNBF5020

Lecture Quick Reference

The seven lectures in one document: what you must be able to do in each lecture, and every exact number Dr. Wong stated.

GNBF5020 · Introduction to Molecular Biology and Genetics

Dr. Tony Wong · Lectures 1-7, 12 Sep - 7 Nov 2026

[ENGLISH ONLY] THE EXAM MUST BE ANSWERED IN ENGLISH. Dr. Wong said this three separate times, and once he said he gave zero marks for it. This document is in English so you can revise directly in the language you will be graded in. The Chinese companion exists only for explanation; gene, protein, pathway, method and drug names stay in English in both versions.

[HOW TO READ THIS] One page per lecture, for the whole course. The Beginner's Guide teaches the content from zero; the Midterm Drill Pack trains the exam format; this is the lookup sheet you keep beside you.

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Lecture 1 - Introduction to Biomolecules - 12 Sep 2026

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Lecture 5 - Control of Gene Expression - 17 Oct 2026

Lecture 6 - Methods in Molecular Biology - 31 Oct 2026

Lecture 7 - Genomic Applications in Biomedical Research - 7 Nov 2026

Quick self-test

Lecture 1 - Introduction to Biomolecules

You must be able to:

- State the central dogma and name its exceptions (reverse transcriptase, RNA replication, prions)
- Distinguish the four biomolecule classes and give one function of each
- Draw a nucleotide and a peptide: label 1'/3'/5', phosphate, sugar, base
- Explain why DNA uses thymine rather than uracil
- Describe primary -> secondary -> tertiary -> quaternary structure
- Explain allostery, and why a chaperone matters
- Explain why hydrogen bonds and base stacking both matter for DNA stability
- Explain the Warburg effect, and name the platforms used to profile metabolites
- Explain what a predictive biomarker is, using KRAS G12C and cetuximab in colorectal cancer

The numbers and facts he gave

Item	What it is	Why it matters
4	The four biomolecule classes: carbohydrate, lipid, protein, nucleic acid	The skeleton of the whole course
3	Nucleotide components: phosphate, sugar, base	Draw it correctly
2	Hydrogen bonds in A-T; 3 in G == C	DNA structure questions
20	Standard amino acids	Every MCQ with an amino acid
1.65 turns / 147 bp	DNA wrapped per nucleosome	Chromatin structure
64	The codons in the standard genetic code, from 4 bases in 3 positions	Read the codon table
pyruvate -> lactate	The Warburg switch: cancer cells convert pyruvate to lactate even with O2 present	Cancer metabolism
13 % / 3 %	KRAS G12C in lung adenocarcinoma / in colorectal cancer	The predictive-biomarker example
13.2 vs 6.7 months	Median overall survival with daraxonrasib vs. standard chemotherapy in metastatic pancreatic cancer	The drug-development example
>= 200 nt	RNA length above which it is a long non-coding RNA (lncRNA)	Lect 5
22 nt	Length of a microRNA	Lect 5

Common traps

- Lipids are not polymers.
- Peptide bonds join amino acids; phosphodiester bonds join nucleotides; glycosidic bonds join sugars.
- Unambiguous != specific. The code is degenerate and unambiguous.
- Higher GC content means a higher melting temperature (Tm). That is the direction he wants, and it is what you apply in PCR primer design.
- The Warburg effect is aerobic glycolysis: pyruvate -> lactate even when oxygen is present.
- Tumour-adjacent tissue is not normal tissue. It is biochemically different, so never treat it as "healthy" in a biomarker comparison.

Lecture 2 - Basic Cell Biology

You must be able to:

- Compare prokaryote and eukaryote, with the endosymbiotic evidence
- Name the organelles and say what each does
- Draw the fluid mosaic model and label phospholipid bilayer, protein, carbohydrate, cholesterol
- Distinguish simple diffusion, facilitated diffusion, osmosis, and active transport
- Predict what happens to a cell in a hypertonic, hypotonic, or isotonic solution
- Lay out G1 -> S -> G2 -> M -> cytokinesis and say what happens in each
- Distinguish mitosis from meiosis I, and say what crossing over and independent assortment do
- Read a pedigree and identify the inheritance pattern

- Explain how flow cytometry separates cells, and what forward scatter and side scatter measure

The numbers and facts he gave

Item	What it is	Why it matters
0.1-0.4 um	*E. coli* size	Prokaryote scale
20-30 um	Typical mammalian cell	Eukaryote scale
2 membranes	Mitochondria and chloroplasts	Endosymbiotic evidence
70S / 80S	Prokaryote / eukaryote ribosome	Why some antibiotics are safe
23 pairs (46)	Human chromosomes	Genome structure
V ~ 39, D = 27, J = 6	V(D)J gene segment copy numbers	Lect 4, and SCID
3	The hallmarks of mitochondrial inheritance: an affected mother passes it to all her children; an affected father passes it to none; only females transmit it onward	Read a pedigree
0.5-1 %	FBS in serum starvation, used to arrest cells in G1	Cell-cycle synchronisation
3 germ layers	Ectoderm, mesoderm, endoderm	The teratoma assay that validates an iPSC
4	Yamanaka factors: Oct3/4, Sox2, Klf4, c-Myc	iPSC reprogramming

Common traps

- The trap he flagged: mitochondrial disease can be caused by a mutation in nuclear DNA too, and then it inherits in a Mendelian way. Do not read "mitochondrial" off the pedigree alone - ask which gene.
- M1 versus M2 non-disjunction: in M1 the homologous chromosomes fail to separate, so one meiosis gives 2 normal and 2 abnormal gametes of the same type; in M2 the sister chromatids fail to separate, giving 1 normal and 3 abnormal.
- The threshold effect: how severe a mitochondrial disease is depends on the ratio of mutant to wild-type mtDNA in a tissue exceeding that tissue's tolerance.
- Cytokinesis partitions organelles at random, which is why siblings carrying the same mitochondrial mutation can differ in severity.
- IgG1 engages Fc receptors on NK cells and triggers ADCC; IgG4 does not, but it has the longer serum half-life.
- Receptor occupancy: about 2 % in breast cancer versus about 10 % in gastric cancer for HER2.

Lecture 3 - DNA and Chromatin Structure

You must be able to:

- State the experiments proving DNA is the genetic material (Griffith, Avery, Hershey-Chase)
- Describe Watson-Crick: antiparallel, complementary, base pairs, grooves
- Draw a nucleotide, a phosphodiester linkage, and the double helix
- Define nucleosome, chromatin, euchromatin, heterochromatin, and the packing levels
- Explain what a histone mark does, and distinguish activating from repressive marks

- Define centromere, telomere, and origin of replication
- Explain methylation and how it is measured (bisulfite sequencing)
- Say which methyltransferase does de novo methylation and which maintains it in the daughter strand
- Read a G-banded karyogram: the chromosome pairs, the p and q arms, and the resolution limit

The numbers and facts he gave

Item	What it is	Why it matters
1953	Watson and Crick	The model
1958	Meselson and Stahl	Proved semi-conservative replication
1962	Nobel Prize (Watson, Crick, Wilkins)	Context
147 bp	DNA per nucleosome core	Chromatin
1.65 turns	Turns of DNA per nucleosome	Chromatin
2.8 A	Resolution of the nucleosome core crystal structure	Nucleosome structure
0.34 nm / 10 bp	Base-pair spacing; one helical turn is 3.4 nm	DNA structure
2 -> 11 -> 30 -> 300 -> 700 -> 1,400 nm	The levels of DNA packing; 30 nm is the (classical) solenoid fibre	Packing levels
3-20 kb	Typical telomere length, in TTAGGG repeats	Chromosome ends
50-200 bp	Telomere lost at each somatic division	End-replication problem
200 nt	RNA length above which it is an lncRNA	Lect 5
425 and 250 bp	The two bands in the liver-cancer splicing gel	Lect 5

Common traps

- Antiparallel, not parallel. This one answer explains leading vs. lagging.
- Major groove is the wider one, and it is where sequence-specific proteins read DNA.
- Promoter methylation represses; demethylation activates.
- Bisulfite converts unmethylated C to uracil and leaves methylated C alone. The reverse is the classic wrong answer.
- The nucleosome core is 147 bp of DNA, not 146; and H1 is the linker histone sitting at the DNA entry and exit, not a fifth subunit of the octamer.
- q21.1 is read "q-two-one-point-one", not "q-twenty-one-point-one".

Lecture 4 - DNA Replication, Repair and Recombination

You must be able to:

- State the Meselson-Stahl result after one and two generations
- Name every enzyme at the replication fork and say what it does
- Explain why the leading strand is continuous and the lagging strand is not
- Describe the end-replication problem and how telomerase solves it
- Name all six repair pathways and match each to the lesion it handles

- Explain why NHEJ failure causes SCID
- Explain homologous recombination, site-specific recombination, and transposition
- Identify mutagens by name and say what lesion each produces
- Explain how a single-nucleotide change and a frameshift each change the protein

The numbers and facts he gave

Item	What it is	Why it matters
30 %	Midterm's share of the final grade	Lect 1, but plan the midterm around it
V ~ 39, D = 27, J = 6	V(D)J segment copy numbers	SCID chain
39 / 27 / 6	The raw material for immune repertoire diversity	SCID chain
15 %	Share of colorectal cancers that are MSI-high	MMR failure -> hot tumour
3 or more	HBV integration sites -> significantly poorer survival	Lect 2
Top 10 %	Longest telomeres in hepatocellular carcinoma had worse recurrence-free survival	Counter-intuitive; worth remembering
39.6	Odds ratio, CIMP-high vs CIMP-0 (95 % CI 20.6-76.1, $p < 0.0001$)	Lect 3
12.7	Odds ratio for detected methylation (CI 7.84-20.7, $p < 0.0001$)	Lect 3
~200 nt / ~10 nt	Okazaki primer spacing, and primer length	The lagging strand
4	The mechanisms that generate a copy-number variation: NAHR, FoSTeS, NHEJ, MEI	Lect 3, CNV

The six repair pathways

Pathway	Lesion it handles	Disease when defective
Direct reversal	One specific chemical lesion (photolyase for UV dimers; MGMT for O6-methylguanine)	- (humans lack photolyase)
BER	A single damaged base	MUTYH-associated polyposis
NER	Bulky lesions spanning several nucleotides	Xeroderma pigmentosum
MMR	Replication errors, small insertion-deletion loops	Lynch syndrome (HNPCC)
HR	Double-strand breaks, with a homologous template	BRCA1/BRCA2 -> breast, ovarian, prostate
NHEJ	Double-strand breaks, no template	Some SCID

Common traps

- BER vs NER: one base versus a whole patch. NER failure gives you sunburn - that is your discriminator.
- HR vs NHEJ: accurate with a template versus fast and error-prone without one.
- UV makes pyrimidine dimers. EMS alkylates (G -> A transitions). Ethidium bromide intercalates (frameshifts). Aflatoxin B1 gives G -> T transversions.
- Deamination of 5-methylcytosine -> thymine is silent and unrepairable, because it looks like a normal base.
- Humans have no photolyase, so a UV dimer is not directly reversed in us - it is removed by NER.

- CRISPR borrows NHEJ: Cas9 cuts, NHEJ repairs the double-strand break well enough to make indels, and the frameshift destroys the gene. The same pathway you met here is the tool in Lectures 6 and 7.

Lecture 5 - Control of Gene Expression

You must be able to:

- Explain the lac operon with both conditions and name the molecule responsible for each
- Explain the trp operon as the mirror image, and define corepressor
- Define cis and trans, and give the chromosome 1 / chromosome 13 example
- List the levels at which eukaryotic gene expression is controlled
- Explain alternative splicing and why 95 % of human multi-exon genes use it
- Explain epigenetic memory and how a differentiated cell stays differentiated
- Explain microRNA, lncRNA, and the difference between them
- Explain codon optimisation and why CHO cells need it
- State the NF-kappaB cytokine time course

The numbers and facts he gave

Item	What it is	Why it matters
2 conditions	Lactose present *and* glucose low	The classic wrong answer is naming only lactose
3 genes	lacZ, lacY, lacA	One polycistronic message
5 genes	trpE, trpD, trpC, trpB, trpA	Switched off when Trp is plentiful
100 uM	Optimal IPTG concentration (range tested 0-1000 uM)	Recombinant production
200 nt	Length above which an RNA is an lncRNA	vs. small RNA
22 nt	Length of a microRNA	Binds a seed sequence
425 and 250 bp	Two bands in the liver-cancer splicing gel	Alternative splicing
23 and 19	Two patient groups: without vs. with alternative splicing	P < 0.001 on the log-rank test
220 and 97	microRNAs screened, and paired tumour / adjacent non-tumour tissues	Liver cancer screen
14q32.2	The over-expressed microRNA cluster	P = 4.23 x 10^-10 vs log10 AFP; survival P = 0.009, chi2 = 6.81
Chromosome 1 / 13	A repressor on chr 1 acting on a gene on chr 13	The clearest trans-regulation example
4-6 hours	When TNF-alpha peaks	First cytokine of the NF-kappaB response
Order	TNF-alpha -> IL-1beta -> IFN-gamma -> IL-2 and IL-15	Know the shape and the time course

The lac operon, precisely

Condition	Mechanism	Molecule
Lactose present	Allolactose binds the Lac repressor and removes it from the operator	Allolactose (made by lacZ-encoded beta-galactosidase - not lactose itself)
Glucose low	cAMP rises; the cAMP-CAP complex binds the promoter and recruits the copying machinery	cAMP-CAP

The trp operon is the mirror image: tryptophan binds the Trp repressor as a corepressor, the complex binds the operator, and tryptophan biosynthesis is switched off. IPTG is the artificial, non-metabolised inducer used in *E. coli* recombinant systems.

Common traps

- Naming only "lactose" for the lac operon - half marks at best.
- Calling a repressor cis and a factor trans backwards: cis = DNA on the same molecule; trans = a diffusible factor.
- NF-kappaB: the cytokine list is not the exam point; the sequence and the time course are.
- Allolactose, not lactose, is what binds the Lac repressor. Lactose is hydrolysed by beta-galactosidase; IPTG is the artificial inducer that is never metabolised at all.
- The miRNA seed sequence is positions 2-8, and G:U wobble pairs are tolerated - it is a partial match, not a perfect one.
- Codon optimisation buys yield, not folding. Delaying induction until exponential phase is what prevents misfolding and inclusion bodies.

Lecture 6 - Methods in Molecular Biology

You must be able to:

- Explain why the sample, not the technique, comes first: FFPE vs fresh frozen, serum vs plasma, bulk tissue vs single cells
- Explain real-time PCR: the principle, the Ct value, and why a larger Ct means fewer copies
- Distinguish SYBR Green from TaqMan, and say what makes TaqMan specific
- Distinguish microarray, RNA-Seq and scRNA-Seq by the question each one answers
- Explain ChIP-Seq, and say why it must be validated with a wet-lab reporter assay
- Separate proteins with SDS-PAGE, isoelectric focusing and 2D-PAGE, and say what each separates on
- Identify proteins by mass spectrometry, and say why LC-MS cannot find a novel protein but LC-MS/MS can
- Explain a positive and a negative CRISPR screen, and state what the untreated control arm is for
- Choose the method: given a question, name the technique, give the reason, and say what it still cannot do

The numbers and facts he gave

Item	What it is	Why it matters
28S and 18S	The two rRNA peaks that make an intact human RNA sample look good on a Bioanalyzer	Sample quality
2 : 1	The expected 28S : 18S ratio in intact mammalian RNA	A 1 : 1 ratio means degradation
RIN < 6	The slide's own cut-off for poor quality (RIN 10 intact, RIN 5 partial, RIN 3 strongly degraded)	The RNA quality gate
DV200	The > 200 nt fraction - the FFPE metric, because RIN loses the ribosomal peaks	FFPE samples
> 95 %	The share of total cellular RNA the slide attributes to rRNA	Must be depleted before library prep
Ct ~13-17	The Ct band quoted for the house-keeping genes used to normalise real-time PCR	Larger Ct = fewer copies
2	The components that make TaqMan specific: a primer *and* a probe	Specificity
1014 vs 3139	DEGs for ANIT on microarray vs RNA-Seq (APAP 92 vs 512)	Why RNA-Seq replaced the array
100 / 1,000 / 10,000	How many genes a microarray measures simultaneously	Microarray scale
21 / 316 / 397,810	Cancer types / patients / T cells in the pan-cancer tumour-infiltrating T-cell atlas	scRNA-Seq
5-250 kDa; 150 / 50 / 25 kDa	The SDS-PAGE size range; intact IgG, heavy chain, light chain	Reducing vs non-reducing
10-20	How many proteins a multiplex assay measures in a single sample	Immunoassays

Common traps

- Positive versus negative CRISPR screen: in a positive screen the knockout is enriched among survivors, so the gene confers drug sensitivity; in a negative screen the knockout is depleted because the cells die, so the gene confers drug resistance. Always state the untreated control arm.
- NHEJ, not MMR - and say both limbs: defective non-homologous end joining means no V(D)J recombination, so no TCR repertoire and no antibody repertoire, and therefore SCID.
- Describe versus explain: when the verb is "explain", naming the mechanism is mandatory.
- SYBR Green binds any double-stranded DNA, so it is not sequence-specific and it detects primer-dimers; TaqMan needs a target-specific probe with a reporter and a quencher.
- Lipid transfection fails on primary T cells, NK cells and macrophages - use electroporation there.
- LC-MS gives a database fingerprint, so it cannot identify a novel protein; LC-MS/MS fragments to two-amino-acid fragments and performs de novo sequencing.

Lecture 7 - Genomic Applications in Biomedical Research

You must be able to:

- Draw the pipeline: omics data -> pathway -> target -> drug -> test -> score -> decision -> patient
- Explain driver versus passenger, and say which two lines of evidence you need

- Distinguish a biomarker from a companion diagnostic (CDx), and say how the CDx is developed
- Define prognostic and predictive, and say what proof each one needs
- Distinguish MammaPrint from Oncotype DX: genes, platform, output, risk bands, and what each one may claim
- Explain the analytical validation, clinical validity and clinical utility ladder
- Explain qPCR versus ddPCR for a rare mutation in ctDNA
- Explain the miRNA classifier for early HCC, and say what a longitudinal serum sample proves
- Explain the biomarker-development block he flagged: discovery, validation, utility, trial design, and the enrichment versus all-comer design

The numbers and facts he gave

Item	What it is	Why it matters
70 vs 21	MammaPrint: 70 genes on a microarray. Oncotype DX: 21 genes (16 cancer + 5 reference) on RT-qPCR	The two breast-cancer signatures
3	The Recurrence Score risk bands: < 18 low, 18 to < 31 intermediate, 31 to 100 high	Where the patient falls
2.81 (95 % CI 1.70-4.64), P < 0.001	The Recurrence Score hazard ratio **per 50-point increment** of RS, not per single point	The only factor whose CI excludes 1, apart from poorly differentiated grade
447	The derivation cohorts, three studies in total, including the tamoxifen arm of NSABP B-20	Where the score came from
668 of 675	Archival NSABP B-14 blocks that were analysable	The prospective validation
295	The independent validation cohort for the 70-gene signature (van de Vijver et al., NEJM 2002)	MammaPrint's validation
651 / 227 / 424	NSABP B-20: patients assessable; tamoxifen alone; tamoxifen plus chemotherapy	The randomised evidence
0.038	The treatment-by-biomarker interaction test in that analysis (Paik et al., J Clin Oncol 2006)	What makes the score predictive
10 / 5 / 2.5 / 1.25 ng	The four standard-curve points for absolute quantification by real-time PCR, across 4 tubes	qPCR vs ddPCR
96 / 15 / 30 / 29 / 57	miRNA discovery set, longitudinal serum set, and validation cohorts (miR-15b + miR-130b)	The miRNA classifier
0.981	AUC of that classifier (sensitivity 98.25 %, specificity 91.53 %, accuracy 94.83 %)	The ROC result
2 cm	The distance from the tumour edge at which cancer-adjacent tissue was sampled	His closing message

Corrections to the lecture deck - write these, not the slides

What the slide says	What the source says	What to write in the exam
RS hazard ratio 2.81 (1.70-4.64), P < 0.001, read as per point	Paik et al. 2004 computed it "relative to an increment of 50 units" of the score	HR 2.81 (1.70-4.64) **per 50-point increment** of RS, P < 0.001
Developed from NSABP B-14, validated with NSABP B-20	Inverted	Derivation: **B-20** plus two other cohorts, 447 patients. Prospective validation: archival **B-14**, 668 of 675 analysable
"Sparano et al., NEJM, 2005"	No such paper exists	**Paik et al., J Clin Oncol 2006**;24(23):3726-3734 - 651 patients, 227 tamoxifen alone, 424 tamoxifen + chemotherapy, interaction test P = 0.038

What the slide says	What the source says	What to write in the exam
Adjacent-tissue slide cited "Badiat-Dekic et al., 2013"	No such author in PubMed	**Roman-Perez et al., Breast Cancer Res 2012;14:R51**
70-gene validation (n = 295) cited "Desmedt et al., Int J Cancer, 2007"	Desmedt 2007 is the **76-gene** signature, n = 198	**van de Vijver et al., NEJM 2002** for the 295-patient validation
21-gene panel lists "ST3 (MMP-11)" and "CATL2"	ST3 is the protein alias of MMP11; the gene is CTSL2	**MMP11** (protein alias ST3) and **CTSL2**, both in the invasion group
Abrocitinib as NDA 213878	213878 is a different application	**NDA 213871**; CYP2C19 poor metabolisers start at 50 mg once daily, escalating to 100 mg after 12 weeks
BEST attributed to NCI	**BEST (Biomarkers, EndpointS, and other Tools)** is a joint FDA-NIH resource (NBK326791)	Attribute it to the **FDA-NIH Biomarker Working Group**

Common traps

- The biomarker is not the companion diagnostic: HER2 is the biomarker, HER2 IHC or HER2 qPCR is the CDx, and the CDx is developed alongside the drug.
- Prognostic is not predictive: prognosis is the outcome whatever treatment you give, prediction is differential benefit from this treatment, and a predictive claim needs a treatment arm.
- MammaPrint is prognostic (70 genes, microarray, 2 bands); Oncotype DX is both prognostic and predictive (21 genes, RT-qPCR, 3 bands). A prognostic assay alone cannot justify withholding a treatment.
- Analytical validation, clinical validity, clinical utility: does the assay measure its target reproducibly, is it correlated with the endpoint, and does using it improve the outcome. A perfect assay with no utility changes nothing.
- The adjacent-tissue paper is Roman-Perez et al., Breast Cancer Res 2012;14:R51. The slide's "Badiat-Dekic et al., 2013" does not exist. In the exam say what he taught; keep the paper separately.
- The 21-gene panel is MMP11 (its protein alias is ST3) and CTSL2 - not "ST3" and "CATL2". BEST is an FDA-NIH resource, not NCI, and abrocitinib is NDA 213871. The n = 295 validation of the 70-gene signature is van de Vijver et al., NEJM 2002; Desmedt et al. 2007 is the 76-gene signature, n = 198.

Quick self-test

Tick a box only if you can do it today, unaided.

- [ ] I can write the central dogma and name three exceptions.
- [ ] I can label a nucleotide and a peptide.
- [ ] I can say why DNA uses thymine.
- [ ] I can distinguish the four biomolecule classes and their functions.
- [ ] I can name the organelles and their jobs.
- [ ] I can predict a cell's fate in a hypertonic solution and say why.
- [ ] I can lay out the cell cycle and the stages of mitosis.
- [ ] I can name the three hallmarks of mitochondrial inheritance and the nuclear-DNA trap.
- [ ] I can draw a nucleosome and describe the packing levels.
- [ ] I can name the Meselson-Stahl result and every enzyme at the fork.
- [ ] I can explain why the lagging strand must be discontinuous.
- [ ] I can match all six repair pathways to their lesions.

- [ ] I can draw the NHEJ -> V(D)J -> SCID chain with the copy numbers.
- [ ] I can state both lac operon conditions and the molecule for each.
- [ ] I can name the three lac genes and the five trp genes.
- [ ] I can define cis and trans with the chromosome 1 / 13 example.
- [ ] I can explain alternative splicing with the 425 / 250 bp bands.
- [ ] I can state the microRNA length, the lncRNA cutoff, and the IPTG optimum.
- [ ] I can say why CHO cells need codon optimisation.
- [ ] I can state the NF-kappaB cytokine order and the TNF-alpha peak time.
- [ ] I can say why the sample, not the technique, comes first in every method.
- [ ] I can tell SYBR Green from TaqMan, and say why a larger Ct means fewer copies.
- [ ] I can separate microarray, RNA-Seq and scRNA-Seq by the question each one answers.
- [ ] I can explain a positive CRISPR screen and a negative one, and say what the untreated control arm is for.
- [ ] I can tell a biomarker from its companion diagnostic, and tell prognostic from predictive.